



Written examination: 21.12.2025 (15:00 - 19:00)
Course number: 02452
Course name: Machine Learning
Aids allowed: No aids, except two A4 sheets with handwritten notes (front and back) and standard dictionaries.
Exam duration: 4 hours.
Format: The exam is multiple choice. All questions have four possible answers marked by the letters A, B, C, and D as well as the answer “Don’t know” marked by the letter E. A correct answer gives 3 points, a wrong answer gives -1 point, and “Don’t know” (E) gives 0 points.
Weighting: Exam questions are weighted equally.
Answers: This exam only allows for physical hand-in. Select **one** answer per exam question by completely filling in a circle (e.g. ☐ A ☐ B ☒ C ☐ D ☐ E) using a black/dark pencil (do **not** use a pen). Do **not** use checkmarks or crosses to indicate your desired answer. To change an answer erase the previously marked answer using an eraser and fill another circle leaving only one clearly selected answer. You **must** answer E if you do not wish to provide an answer for a question.

Building/room:
(required)

Table no.:
(required)

Student ID:
(required)

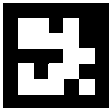
s						
---	--	--	--	--	--	--

Legibly write your student ID (without the s) e.g. 123456 in the form and fill in the corresponding circles below the handwritten digits (required).

☐ 0	☐ 0	☐ 0	☐ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1	☐ 1	☐ 1	☐ 1
☐ 2	☐ 2	☐ 2	☐ 2	☐ 2	☐ 2
☐ 3	☐ 3	☐ 3	☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9	☐ 9	☐ 9	☐ 9



	A	B	C	D	E
Question 1	☐ A	☐ B	☐ C	☐ D	☐ E
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E
Question 4	☐ A	☐ B	☐ C	☐ D	☐ E
Question 5	☐ A	☐ B	☐ C	☐ D	☐ E
Question 6	☐ A	☐ B	☐ C	☐ D	☐ E
Question 7	☐ A	☐ B	☐ C	☐ D	☐ E
Question 8	☐ A	☐ B	☐ C	☐ D	☐ E
Question 9	☐ A	☐ B	☐ C	☐ D	☐ E
Question 10	☐ A	☐ B	☐ C	☐ D	☐ E
Question 11	☐ A	☐ B	☐ C	☐ D	☐ E
Question 12	☐ A	☐ B	☐ C	☐ D	☐ E
Question 13	☐ A	☐ B	☐ C	☐ D	☐ E
Question 14	☐ A	☐ B	☐ C	☐ D	☐ E
Question 15	☐ A	☐ B	☐ C	☐ D	☐ E
Question 16	☐ A	☐ B	☐ C	☐ D	☐ E
Question 17	☐ A	☐ B	☐ C	☐ D	☐ E
Question 18	☐ A	☐ B	☐ C	☐ D	☐ E
Question 19	☐ A	☐ B	☐ C	☐ D	☐ E
Question 20	☐ A	☐ B	☐ C	☐ D	☐ E
Question 21	☐ A	☐ B	☐ C	☐ D	☐ E
Question 22	☐ A	☐ B	☐ C	☐ D	☐ E
Question 23	☐ A	☐ B	☐ C	☐ D	☐ E
Question 24	☐ A	☐ B	☐ C	☐ D	☐ E
Question 25	☐ A	☐ B	☐ C	☐ D	☐ E
Question 26	☐ A	☐ B	☐ C	☐ D	☐ E
Question 27	☐ A	☐ B	☐ C	☐ D	☐ E



THIS PAGE IS INTENTIONALLY LEFT BLANK

	Attribute description	Abbrev.
x_1	The weight of the fish (in grams).	Weight
x_2	The length of the fish (in cm).	Length
x_3	The height of the fish (in cm).	Height
x_4	The width of the fish (in cm).	Width
y	The type of the fish.	Fish type

Table 1: Description of the attributes of the Fish market dataset used in this exam. The Fish market dataset is a collection of data related to various species of fish and their characteristics. We can estimate the type of a fish (Small fish vs. Big fish) based on the provided physical measurements.

Question 1. We standardize the attributes of the Fish market dataset. We then consider the histograms of the four attributes in Figure 2. Match the histograms to the attributes by using the information regarding the distributions of the points from Figure 1. Which one of the following statements is *correct*?

- A. Histogram 1: x_4 , Histogram 2: x_1
Histogram 3: x_2 , Histogram 4: x_3 .
- B. Histogram 1: x_3 , Histogram 2: x_4
Histogram 3: x_1 , Histogram 4: x_2 .
- C. Histogram 1: x_4 , Histogram 2: x_2
Histogram 3: x_1 , Histogram 4: x_3 .
- D. Histogram 1: x_2 , Histogram 2: x_4
Histogram 3: x_1 , Histogram 4: x_3 .
- E. Don't know.

Question 2. Consider a small dataset comprised of $N = 7$ one-dimensional observations shown as the filled circles in Figure 3.

Suppose a k -means algorithm is applied to the dataset with $K = 2$ and using Euclidean distances. We will assume the location of the centroids are initialized to the values indicated by the crosses in Figure 3. After initialization, the k -means algorithm is evaluated for one step, comprised of assigning observations to centroids and updating the location of the centroids. After the first step, what will be the new location of the centroids?

- A. $\mu_1 = \frac{29}{5}$, and $\mu_2 = \frac{13}{2}$.
- B. $\mu_1 = \frac{5}{2}$, and $\mu_2 = \frac{37}{5}$.
- C. $\mu_1 = 3$, and $\mu_2 = \frac{33}{4}$.
- D. $\mu_1 = \frac{15}{4}$, and $\mu_2 = 9$.
- E. Don't know.

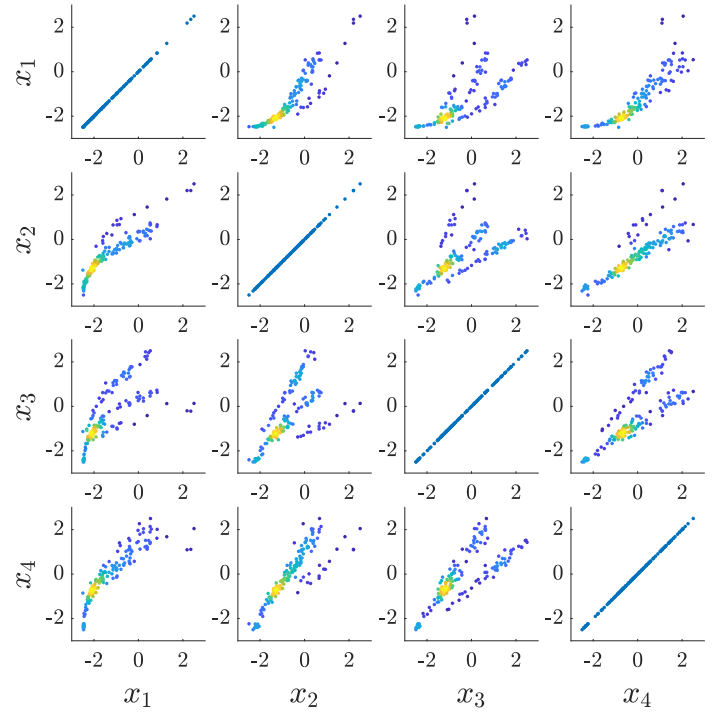


Figure 1: The scatter plot matrix for the attributes x_1 , x_2 , x_3 and x_4 of the Fish market dataset of Table 1. In the off-diagonal panels, the color represents the associated density.

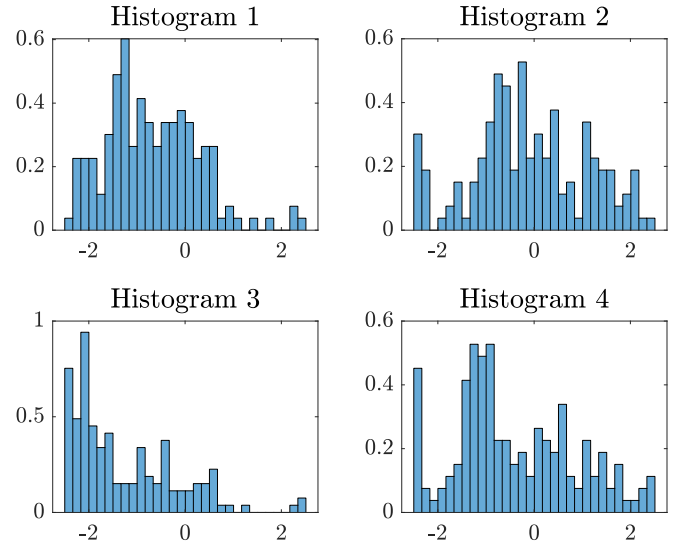


Figure 2: Histograms of the attributes in which their order has been permuted.

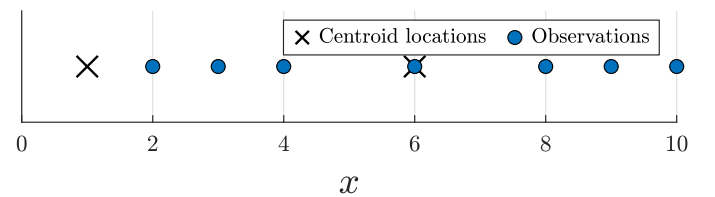


Figure 3: A small 1-dimensional dataset and initial values of centroids.

Question 3. Consider the distances in Table 2 based on 10 observations from the Fish market dataset. The class labels C_1 , C_2 (see table caption for details) will be predicted using a k -nearest neighbour classifier based on the distances given in Table 2 (ties are broken in the usual manner by considering the nearest observation from the tied classes). Suppose we use leave-one-out cross validation (i.e., the observation that is being predicted is left out) and a 3-nearest neighbour classifier (i.e., $k = 3$). What is the accuracy computed for all $N = 10$ observations?

- A. accuracy = 1
- B. accuracy = $\frac{1}{5}$
- C. accuracy = 0
- D. accuracy = $\frac{3}{10}$
- E. Don't know.

Question 4. Which one of the following statements is *correct*?

- A. In a regression problem, an ANN with linear activation functions ($h(x) = x$) and many hidden units results in a complex model that tends to overfit if output depends on the input in a non-linear manner.
- B. Randomly initializing two ANNs with identical activation functions and number of hidden units, and training them with the same data, will *always* converge to the same model.
- C. Regularizing an artificial neural network regression model by penalizing the weights using the squared Euclidean norm and a positive value λ decreases bias and increases variance.
- D. We train a neural network by computing the gradient of the cost $E(\mathbf{w})$ using the chain rule and updating the parameters by moving in the opposite direction with a small step.
- E. Don't know.

Question 5. A hierarchical clustering is applied to the 10 observations in Table 2 using *minimum* linkage. Which one of the dendrograms shown in Figure 4 corresponds to the distances given in Table 2?

- A. Dendrogram 1
- B. Dendrogram 2
- C. Dendrogram 3
- D. Dendrogram 4
- E. Don't know.

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	1.0	0.3	0.8	3.4	6.7	7.8	6.2	5.4	8.8
o_2	1.0	0.0	0.7	1.7	4.2	5.9	7.1	5.4	4.6	8.0
o_3	0.3	0.7	0.0	1.1	3.7	6.4	7.5	5.9	5.1	8.6
o_4	0.8	1.7	1.1	0.0	2.6	7.5	8.5	7.0	6.2	9.6
o_5	3.4	4.2	3.7	2.6	0.0	9.9	10.9	9.4	8.7	11.9
o_6	6.7	5.9	6.4	7.5	9.9	0.0	2.4	1.9	1.6	3.0
o_7	7.8	7.1	7.5	8.5	10.9	2.4	0.0	1.7	3.5	1.3
o_8	6.2	5.4	5.9	7.0	9.4	1.9	1.7	0.0	2.2	2.7
o_9	5.4	4.6	5.1	6.2	8.7	1.6	3.5	2.2	0.0	4.3
o_{10}	8.8	8.0	8.6	9.6	11.9	3.0	1.3	2.7	4.3	0.0

Table 2: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the Fish market dataset (recall that $M = 4$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ corresponding to the attributes in Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5\}$ belong to class C_1 (corresponding to Small fish), and the red observations $\{o_6, o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to Big fish).

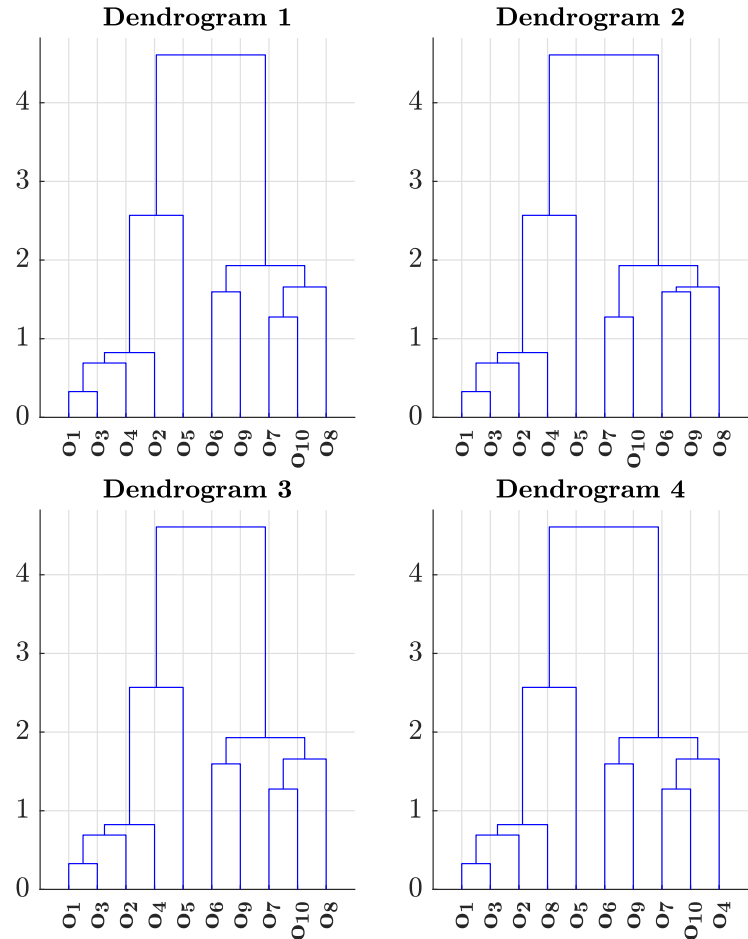


Figure 4: Proposed hierarchical clustering of the 10 observations in Table 2.

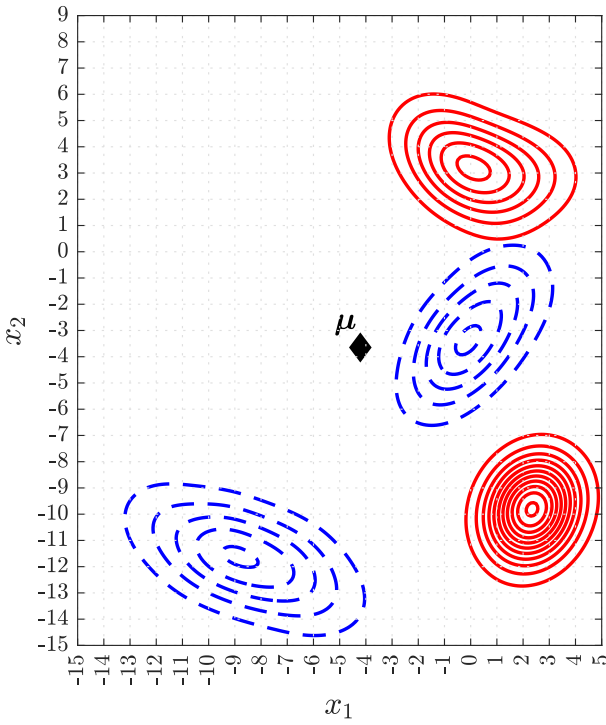


Figure 5: A two-dimensional dataset with two classes, indicated by dashed (--) contours for Class 1 and continuous (—) contours for Class 2. The mean of the dataset is indicated by $\blacklozenge$.

Question 6. Suppose that we perform PCA on a two dimensional dataset presented in Figure 5 where two classes can be distinguished. In Figure 6 the projections of the dataset onto four different directions are presented, one per panel. Two of these plots correspond to the projection on the first and the second principal component.

Which one of the options below is *correct*?

- A. First principal direction corresponds to Plot 1
Second principal direction corresponds to Plot 4
- B. First principal direction corresponds to Plot 3
Second principal direction corresponds to Plot 4
- C. First principal direction corresponds to Plot 1
Second principal direction corresponds to Plot 3
- D. First principal direction corresponds to Plot 4
Second principal direction corresponds to Plot 2
- E. Don't know.

Question 7. Suppose that we perform PCA on a four dimensional dataset with mean $\mu = [3 \ 2 \ 4 \ -2]^\top$, and consider the projection matrix

$$V = \begin{bmatrix} 0.9 & -0.3 & 0.3 & 0.1 \\ 0.1 & -0.3 & -0.3 & -0.9 \\ 0.3 & 0.1 & -0.9 & 0.3 \\ 0.3 & 0.9 & 0.1 & -0.3 \end{bmatrix}$$

computed by SVD. We are given a point $x = [-3 \ -2 \ -1 \ 3]^\top$ and project it onto the principal components 2 and 4.

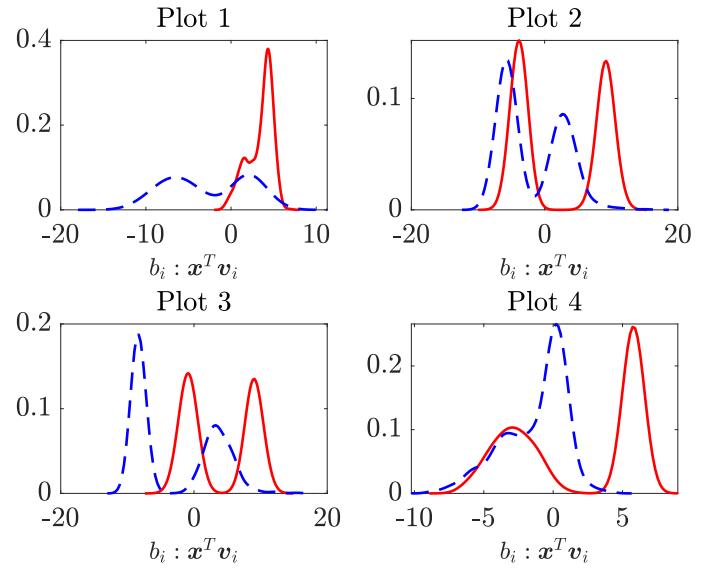


Figure 6: The projection of the data onto four different vectors. Note that each class is represented with a different density in each plot: the Class 1 with continuous (—) contours is represented by the continuous density curve, and the Class 2 with dashed (--) contours is represented by the dashed density curve.

Which one of the following projections is *correct*?

- A. $[b_2, b_4]^\top = [7 \ 0]^\top$
- B. $[b_2, b_4]^\top = [-2 \ -7]^\top$
- C. $[b_2, b_4]^\top = [-2 \ -4]^\top$
- D. $[b_2, b_4]^\top = [4 \ 0]^\top$
- E. Don't know.

Question 8. Which one of the following statements is *correct*?

- A. In a k -Nearest Neighbor classifier, the resulting decision boundaries do not change if we keep the number of neighbors K fixed but change the distance measure between the points.
- B. Decision trees are strictly limited to binary classification tasks.
- C. For every dataset it holds for the eigenvectors computed by SVD that the angle between every pair of distinct eigenvectors is 90° degrees and the Euclidean norm of every individual eigenvector is equal to one.
- D. The number of components K used in the GMM can be determined by selecting the number of components that provides the best log-likelihood of the training data used for training the density model.
- E. Don't know.

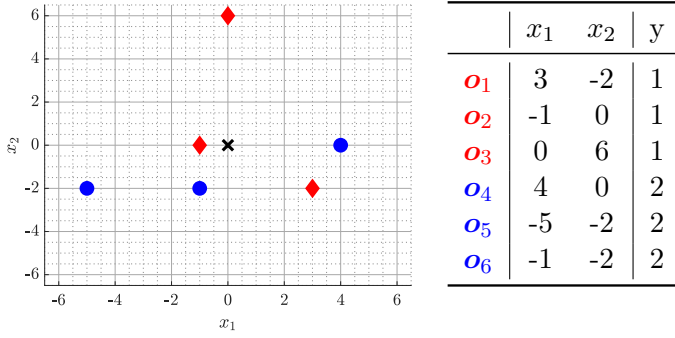


Figure 7: The synthetic dataset, where the shapes indicate the classes such that the observations $\{o_1, o_2, o_3\}$ belong to class C1 ($\diamond$), and the observations $\{o_4, o_5, o_6\}$ belong to class C2 ($\circ$).

Question 9. Consider the data in Figure 7 where shapes indicates classes. We consider the p -distances d_2 , d_∞ , d_1 and the KNN classifier with $K = 3$ to predict the label of the test point $\mathbf{o}_{\text{test}} = [0 \ 0]^\top$.

Which one of the following statements is *correct*?

- A. d_2 predicts C1 , d_∞ predicts C1 , d_1 predicts C1 .
- B. d_2 predicts C1 , d_∞ predicts C1 , d_1 predicts C2 .
- C. d_2 predicts C2 , d_∞ predicts C1 , d_1 predicts C2 .
- D. d_2 predicts C1 , d_∞ predicts C2 , d_1 predicts C2 .
- E. Don't know.

Question 10. Consider a binary classification problem on which two classifiers, Model A and Model B, have been trained to predict if an observation, x , belongs to Class 1 or Class 2. We wish to compare the two models using McNemar's test to determine if there is a significant difference between them, and our null hypothesis is that the classifiers perform similarly. After performing McNemar's test, we obtain:

$$\hat{\theta} = 0.4, \quad [\theta_L, \theta_U] = [-0.02, 0.75], \quad p\text{-value} = 0.22.$$

In Figure 8, the probabilities $p(y = 2 | x_i)$ for the two models for 10 test points $x_1, \dots, x_{10}$ are shown, along with a table providing four options for the true labels of the test points. Only one of the provided labeling options in Figure 8 agrees with the result of the McNemar's test reported above.

Determine which one of the labeling options provided in Figure 8 gives the reported result.

Hint: Generic information about McNemar's test for comparing the accuracy difference $\theta = \theta_A - \theta_B$ of two classifiers.

$$p(\theta | \mathbf{n}) = \frac{1}{2} \text{Beta} \left(\frac{\theta + 1}{2} \mid a = f, b = g \right)$$

where $f = \frac{E_\theta + 1}{2}(Q - 1)$, $g = \frac{1 - E_\theta}{2}(Q - 1)$ and $E_\theta = \frac{n_{12} - n_{21}}{n}$.

$$Q = \frac{n^2(n + 1)(E_\theta + 1)(1 - E_\theta)}{n(n_{12} + n_{21}) - (n_{12} - n_{21})^2}$$

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}
Option 1	1	1	1	2	2	1	2	1	1	1
Option 2	2	1	2	2	2	1	1	2	1	1
Option 3	2	1	1	1	1	2	1	2	2	2
Option 4	1	1	2	1	1	2	2	1	1	1

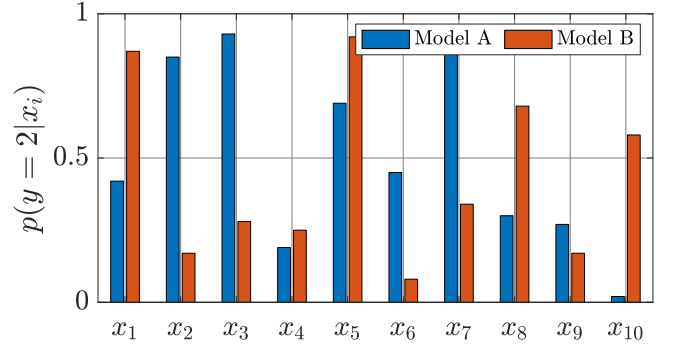


Figure 8: $p(y = 2 | x_i)$ for the two models on 10 points $x_1, \dots, x_{10}$. If $p(y = 2 | x_i) > 0.5$, then x_i is classified as Class 2, otherwise, it is classified as Class 1. For each point, the first bar corresponds to Model A and the second bar to Model B. At the top, the four labeling options are shown, each representing a different assignment of true labels to the test points.

$$\theta_L = 2\text{cdf}_{\text{Beta}}^{-1} \left(\frac{\alpha}{2} \mid a = f, b = g \right) - 1$$

$$\theta_U = 2\text{cdf}_{\text{Beta}}^{-1} \left(1 - \frac{\alpha}{2} \mid a = f, b = g \right) - 1$$

$\mathbf{n}$ is a 2×2 matrix with n_{11} and n_{22} being the number of times model A and model B are simultaneously correct and wrong, respectively. n_{12} is the number of times model A is correct but model B wrong and vice-versa for n_{21} .

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4
- E. Don't know.

Outer fold $i = 1$				
Inner folds	$j = 1$	$j = 2$	$j = 3$	Test
Model 1	19	23	19	6
Model 2	1	15	30	51

Outer fold $i = 2$				
Inner folds	$j = 1$	$j = 2$	$j = 3$	Test
Model 1	8	5	7	3
Model 2	14	3	24	84

Table 3: For each outer fold, we are given the number of misclassified points for each model for each inner fold on the associated validation set, together with the number of misclassified points on the associated test partition of the outer fold.

Question 11. We consider a subset of $N = 180$ points from the Fish market dataset and two different classifiers (Model A and Model B). We apply two-level cross-validation to estimate the generalization error, with 2 outer folds and 3 inner folds. In Table 3 the information for each outer fold, model, and inner fold is given, together with the performance of each model on the test partition of the outer fold.

Which one of the following options is a correct estimate of the generalization error?

- A. $\hat{E}^{\text{gen}} = 0.3$
- B. $\hat{E}^{\text{gen}} = 0.15$
- C. $\hat{E}^{\text{gen}} = 0.05$
- D. $\hat{E}^{\text{gen}} = 0.75$
- E. Don't know.

Question 12. Which one of the following statements is *correct*?

- A. For any model (e.g., linear regression, logistic regression, ANN, etc.) and any regularized cost with a fixed λ , there is *always* a closed form solution for $\mathbf{w}$, obtained by differentiating the objective function and setting the derivative to zero.
- B. The aim of regularization is to reduce the variance of a model by introducing some bias.
- C. A very small error on the training data is in itself evidence of overfitting.
- D. In regularized linear regression, as the regularization parameter λ increases, the norm of the learned weights tends to increase.
- E. Don't know.

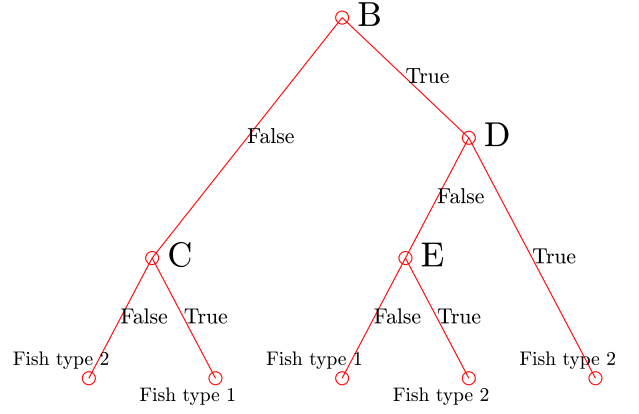


Figure 9: Structure of the decision tree. The goal is to determine the splitting rules.

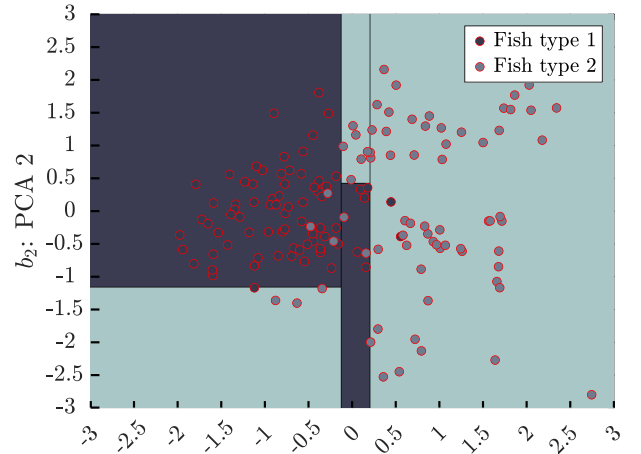


Figure 10: Classification boundary.

Question 13. We consider the Fish market dataset projected onto the first two principal directions. Suppose we train a decision tree to predict which of the two classes an observation belongs to. Since the attributes are continuous, we will consider binary splits of the form $b_i \geq z$ for different values of i and z , where b_1 and b_2 refer to the coordinates of the observations when projected onto principal directions. Suppose the trained decision tree has the form shown in Figure 9, and that according to the tree the predicted label assignment for a subset of the observations is as given in Figure 10, what is then the correct rule assignment to the nodes in the decision tree?

- A. **B:** $b_2 \geq 0.42$, **C:** $b_2 \geq -1.18$, **D:** $b_1 \geq 0.19$, **E:** $b_1 \geq -0.13$
- B. **B:** $b_1 \geq -0.13$, **C:** $b_1 \geq 0.19$, **D:** $b_2 \geq 0.42$, **E:** $b_2 \geq -1.18$
- C. **B:** $b_1 \geq -0.13$, **C:** $b_2 \geq -1.18$, **D:** $b_1 \geq 0.19$, **E:** $b_2 \geq 0.42$
- D. **B:** $b_2 \geq -1.18$, **C:** $b_1 \geq 0.19$, **D:** $b_2 \geq 0.42$, **E:** $b_1 \geq -0.13$
- E. Don't know.

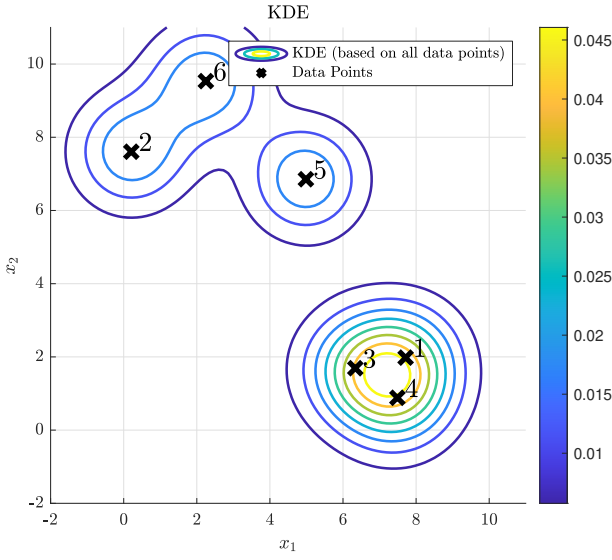


Figure 11: Kernel Density Estimator (KDE)

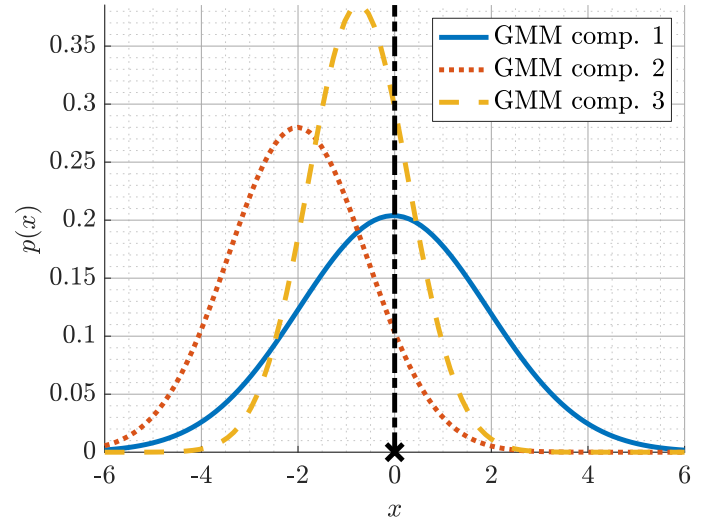


Figure 12: Mixture components in a Gaussian mixture model (GMM) with $K = 3$.

Obs.	Component					
	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$
$\mathbf{x}_1$	0.13	0.00	0.06	0.08	0.00	0.00
$\mathbf{x}_2$	0.00	0.13	0.00	0.00	0.00	0.01
$\mathbf{x}_3$	0.06	0.00	0.13	0.06	0.00	0.00
$\mathbf{x}_4$	0.08	0.00	0.06	0.13	0.00	0.00
$\mathbf{x}_5$	0.00	0.00	0.00	0.00	0.13	0.00
$\mathbf{x}_6$	0.00	0.01	0.00	0.00	0.00	0.13

Table 4: Evaluation of each *individual* component/kernel at each observation. We consider the Gaussian $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \sigma^2 \mathbf{I}_2)$ as the kernel.

Question 14. Figure 11 shows a kernel density estimator (KDE) on 6 observations using the standard formulation of the KDE, i.e.,

$$p(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N \mathcal{N}(\mathbf{x} \mid \mathbf{x}_j, \sigma^2 \mathbf{I}_2).$$

Table 4 shows the evaluation of each *individual* component/kernel at each of the 6 observations. Compute the leave-one-out (LOO) estimate of $p(\mathbf{x}_1)$.

- A. $p(\mathbf{x}_1) = 0.14$
- B. $p(\mathbf{x}_1) = 0.023$
- C. $p(\mathbf{x}_1) = 0.054$
- D. $p(\mathbf{x}_1) = 0.028$
- E. Don't know.

Question 15. We wish to apply the EM algorithm to fit a 1D GMM mixture model to the single feature x from a given dataset. At the first step of the EM algorithm, the $K = 3$ mixture components have densities as indicated by each of the curves in Figure 12 (i.e., each curve is a normalized, Gaussian density $\mathcal{N}(x \mid \mu_k, \sigma_k)$). In the figure, we have indicated the

x -value of a single observation x as a black cross. Suppose we wish to apply the EM algorithm to this mixture model beginning with the *E*-step. We assume the weights of the components are $\mathbf{w} = [0.3 \ 0.6 \ 0.1]$ and the mean/variances of the components are those indicated in the figure.

According to the EM algorithm, what is the probability the observation x is assigned to mixture component 1?

- A. $p(z = 1 \mid x) = 0.4$
- B. $p(z = 1 \mid x) = 0.06$
- C. $p(z = 1 \mid x) = 0.58$
- D. $p(z = 1 \mid x) = 0.2$
- E. Don't know.

Question 16. Which one of the following statements is *correct*?

- A. K -fold cross-validation requires fitting K models in *total* to determine the best among several complexity levels of the model.
- B. The two-layer cross-validation procedure can be used *only* for estimating generalization performance and cannot provide a trained model that can be used in practice, e.g., as an ensemble.
- C. Consider a model class with several complexity levels. The test error obtained by cross-validation and used to select the model complexity is a biased estimate of the generalization error.
- D. For a given dataset, leave-one-out cross-validation typically provides a worse estimate of the generalization error than K -fold cross-validation or hold-out cross-validation.
- E. Don't know.

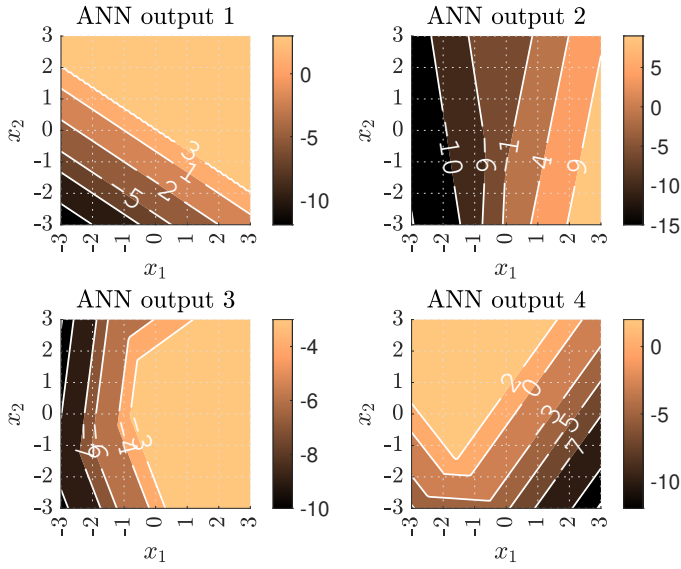


Figure 13: Suggested outputs of an ANN trained on the two attributes x_1 and x_2 from the Fish market dataset to predict the attribute x_3 . Note the signs on the isocurves.

Question 17. Consider an artificial neural network (ANN) trained on the Fish market dataset described in Table 1 to predict the continuous attribute x_3 from the two attributes x_1 and x_2 . Suppose the neural network takes the form:

$$f(x, \mathbf{w}) = h^{(2)} \left(w_0^{(2)} + \sum_{j=1}^2 w_j^{(2)} h^{(1)}([1 \ x_1 \ x_2] \mathbf{w}_j^{(1)}) \right).$$

where the activation functions are selected as $h^{(1)}(x) = \max(x, 0)$ (the rectified linear activation function) and $h^{(2)}(x) = x$ (the linear activation function) and the weights are given as:

$$\mathbf{w}_1^{(1)} = \begin{bmatrix} 0 \\ -2 \\ -3 \end{bmatrix}, \quad \mathbf{w}_2^{(1)} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

$$\mathbf{w}^{(2)} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad w_0^{(2)} = 2.$$

Which one of the functions in Figure 13 will then correspond to the function f ?

- A. ANN output 1
- B. ANN output 2
- C. ANN output 3
- D. ANN output 4
- E. Don't know.

Question 18. Let $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the multivariate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. In Figure 14 we are given four different sets of samples generated from four different Gaussian Mixture Models (GMM) with three components each.

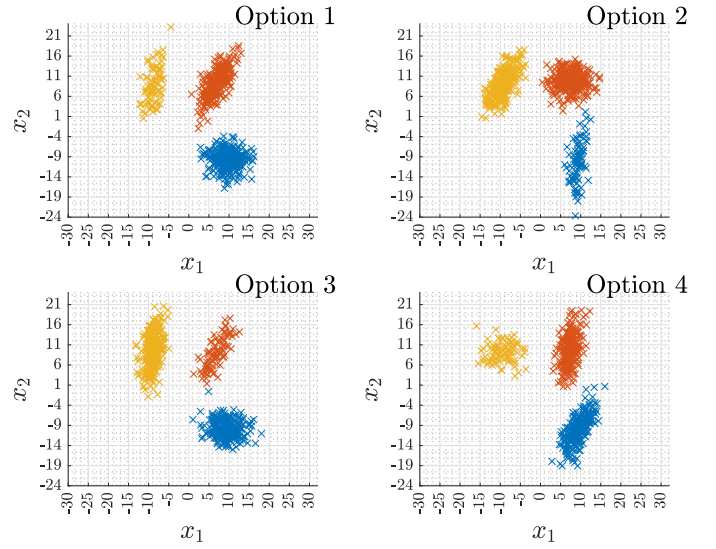


Figure 14: 1000 observations drawn from a Gaussian Mixture Model (GMM) with three clusters.

Each observation is colored according to the component from which it was generated. Consider the GMM:

$$p(\mathbf{x}) = \frac{3}{7} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} 9.5 \\ -9.9 \end{bmatrix}, \begin{bmatrix} 7.5 & 0.1 \\ 0.1 & 5.2 \end{bmatrix} \right) \\ + \frac{3}{7} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} 7.3 \\ 9.7 \end{bmatrix}, \begin{bmatrix} 5.0 & 5.8 \\ 5.8 & 12.9 \end{bmatrix} \right) \\ + \frac{1}{7} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -9.1 \\ 9.1 \end{bmatrix}, \begin{bmatrix} 2.0 & 1.8 \\ 1.8 & 15.5 \end{bmatrix} \right).$$

Which sample set corresponds to this model?

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4
- E. Don't know.

Question 19. Which one of the following statements about ensemble methods is *correct*?

- A. In AdaBoost the more errors a classifier makes in a given round of boosting the less the weights will be increased for the correctly classified points.
- B. In bagging and boosting observations are sampled with replacement, but in bagging misclassified observations are given a higher probability of being sampled in the next round.
- C. In an ensemble method, combining weak classifiers cannot substantially improve performance if the classifiers are independent.
- D. In AdaBoost with a regularized logistic regression classifier as the weak learner in each round of boosting, increasing the number of iterations increases the chance of overfitting.
- E. Don't know.

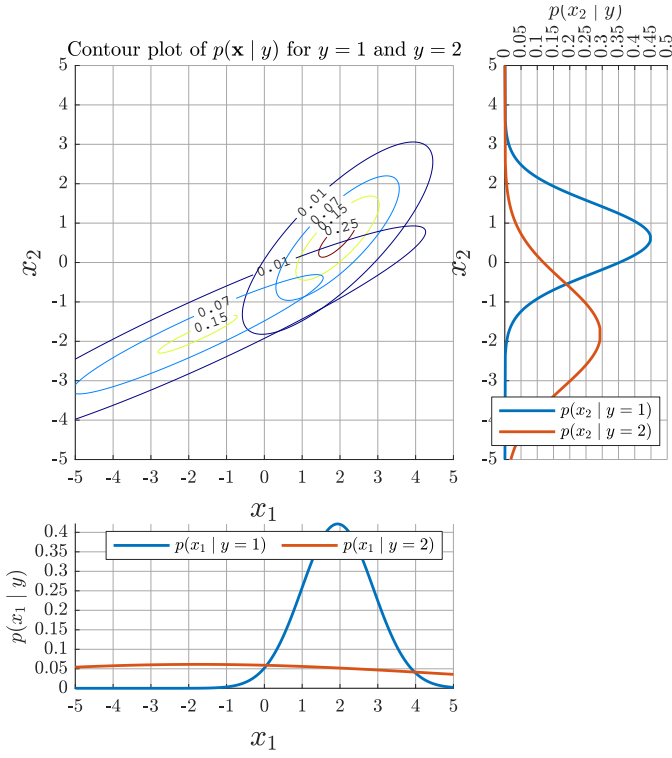


Figure 15: Conditional distributions related to a Naïve Bayes classifier.

Question 20. Consider the task of classifying the label y from two continuous attributes, namely x_1 and x_2 using a Naïve Bayes classifier. We use a normal distribution to model the class conditional distribution of a point $\mathbf{x} = [x_1, x_2]^\top$ and we denote $p(\mathbf{x} \mid \mu_{y=k}, \Sigma_{y=k})$ with $p(\mathbf{x} \mid y = k)$. We consider the point $\mathbf{x} = [1 \ -1]^\top$ and we have that $p(\mathbf{x} \mid y = 1) = 0.02$ and $p(\mathbf{x} \mid y = 2) = 0.02$. The specific distributions are illustrated in Figure 15. Additionally, it is noted that $p(y = 1) = 0.2$ and $p(y = 2) = 0.8$, and a point is classified in class k if $p(y = k \mid \mathbf{x}) \geq 0.5$.

Which one of the following statements is *correct* under the Naïve Bayes classifier?

- A. The point is classified as Class 1.
- B. The point is classified as Class 2.
- C. It holds that $p(y = 1 \mid \mathbf{x}) = p(y = 2 \mid \mathbf{x}) = 0.5$.
- D. It holds that $p(y = 1 \mid \mathbf{x}) = \frac{0.02 \cdot 0.2}{0.02 \cdot 0.2 + 0.02 \cdot 0.8}$.
- E. Don't know.

Question 21. Table 5 shows the confusion matrices related to a Logistic Regression model (LR) and a Decision Tree classifier (DT) trained on a particular instance of the Fish market dataset to predict Fish type. The number of observations is $N = 617$. We define $y = 1$ to indicate a *positive outcome* and $y = 2$ to be a *negative outcome*.

Model A: Logistic Regression (LR)

		Predicted	
		Class 1	Class 2
Actual	Class 1	57	83
	Class 2	296	181

Model B: Decision Tree (DT)

		Predicted	
		Class 1	Class 2
Actual	Class 1	80	60
	Class 2	250	227

Table 5: Confusion matrices.

Which one of the following statements is *correct*?

- A. Model A has higher recall than Model B.
- B. Predicting everything as Class 2 gives worse accuracy than Model A.
- C. Model A has lower error rate than Model B.
- D. Model B has higher precision than Model A.
- E. Don't know.

	x_1	x_2	x_3	x_4	y
$\mathbf{o}_1$	1	0	0	1	1
$\mathbf{o}_2$	1	1	0	1	1
$\mathbf{o}_3$	0	0	0	0	1
$\mathbf{o}_4$	1	0	1	0	1
$\mathbf{o}_5$	1	0	1	1	2
$\mathbf{o}_6$	1	0	1	0	2
$\mathbf{o}_7$	1	1	0	1	2
$\mathbf{o}_8$	0	0	1	1	2

Table 6: The observations $\mathbf{o}_i$ together with the binary attributes x_j and the associated label information y .

Question 22. We consider the dataset in Table 6 with the binary attributes x_j for $j = 1, \dots, 4$. We construct a Bayes classifier relying on the attributes x_1 and x_3 .

Which probability of the following is *correct*?

- A. $p(y = 1 \mid x_1 = 1, x_3 = 1) \approx 0.04$
- B. $p(y = 1 \mid x_1 = 1, x_3 = 1) \approx 0.33$
- C. $p(y = 1 \mid x_1 = 1, x_3 = 1) \approx 0.5$
- D. $p(y = 1 \mid x_1 = 1, x_3 = 1) \approx 0.97$
- E. Don't know.

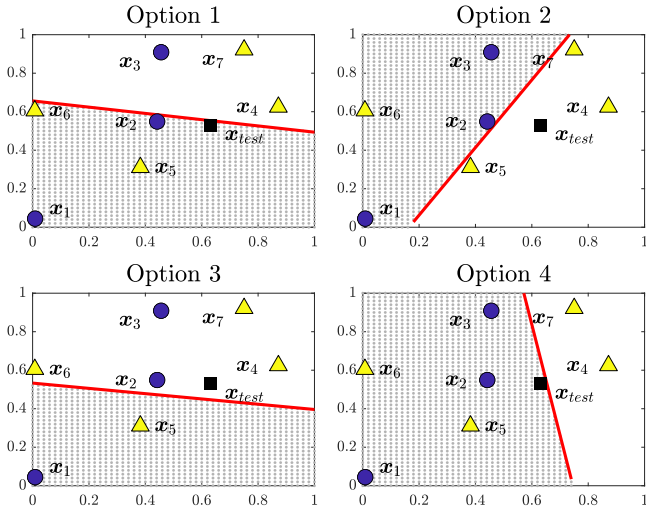


Figure 16: The four possible options for the classifier f_2 . The shaded area is classified as Class 1. The points of Class 1 are represented by $\circ$ and of Class 2 by $\triangle$. The test point is represented by $\square$.

$\mathbf{x}$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$	$\mathbf{x}_6$	$\mathbf{x}_7$
y	1	1	1	2	2	2	2
w_1	0.142	0.142	0.142	0.142	0.142	0.142	0.142
w_2	0.249	0.249	0.1	0.1	0.1	0.1	0.1

Table 7: The observations for the AdaBoost model.

Question 23. We consider the dataset in Figure 16 and the weights in Table 7 after two iterations of the AdaBoost algorithm. The predictions for a test point $\mathbf{x}_{test}$ for the two weak classifiers are $f_1(\mathbf{x}_{test}) = 2$ and $f_2(\mathbf{x}_{test}) = 1$. The final prediction using these two weak classifiers is $f^*(\mathbf{x}_{test}) = 2$.

Which one of the options in Figure 16 corresponds to the classifier f_2 ?

Hint: Consider the weighted error rates of f_1 and f_2 and the AdaBoost prediction rule defined as $f^(\mathbf{x}) = \arg \max_{y=1,2} \sum_{t=1}^T \alpha_t \delta_{f_t(\mathbf{x}), y}$.*

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4
- E. Don't know.

Question 24. A regularized least squares linear regression model is trained using the regularization term $\|\mathbf{w}\|_2^2$ with parameter $\lambda = 3$. We learn the parameters of the model based on the standardized training data $\hat{\mathbf{X}}$, where the mean and standard deviation of the training data are given by $\boldsymbol{\mu} = [4 \ -4]^\top$ and $\boldsymbol{\sigma} = [4 \ 4]^\top$, respectively. The learned parameters are $\mathbf{w}^* = [6 \ 2]^\top$ and $w_0^* = 9$.

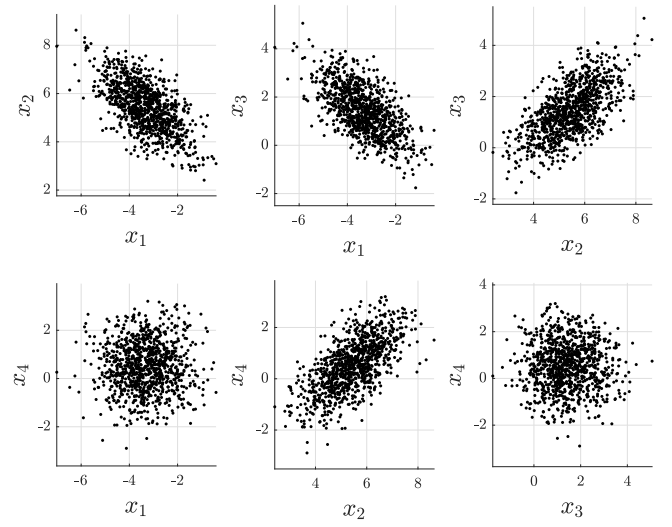


Figure 17: Scatter plot of each pairs of attributes of a vectors $\mathbf{x}$ drawn from a multivariate normal distribution of 4 dimensions.

What is the output of the trained linear regression model for a *new* test point $\mathbf{x}_{test} = [-1 \ -7]^\top$?

- A. $\hat{y} = 120$
- B. $\hat{y} = -11$
- C. $\hat{y} = 0$
- D. $\hat{y} = 109$
- E. Don't know.

Question 25. Consider a multivariate normal distribution with covariance matrix $\boldsymbol{\Sigma}$ and mean $\boldsymbol{\mu}$ and suppose we generate 1000 random samples from it:

$$\mathbf{x} = [x_1 \ x_2 \ x_3 \ x_4]^\top \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Plots of each pair of coordinates of the draws $\mathbf{x}$ is shown in Figure 17.

Which one of the following is the most plausible covariance matrix?

- A. $\boldsymbol{\Sigma} = \begin{bmatrix} 1.0 & -0.65 & 0.65 & 0.65 \\ -0.65 & 1.0 & -0.65 & 0.0 \\ 0.65 & -0.65 & 1.0 & 0.0 \\ 0.65 & 0.0 & 0.0 & 1.0 \end{bmatrix}$
- B. $\boldsymbol{\Sigma} = \begin{bmatrix} 1.0 & -0.65 & 0.0 & -0.65 \\ -0.65 & 1.0 & 0.65 & 0.65 \\ 0.0 & 0.65 & 1.0 & 0.0 \\ -0.65 & 0.65 & 0.0 & 1.0 \end{bmatrix}$
- C. $\boldsymbol{\Sigma} = \begin{bmatrix} 1.0 & 0.0 & 0.65 & 0.0 \\ 0.0 & 1.0 & -0.65 & -0.65 \\ 0.65 & -0.65 & 1.0 & 0.65 \\ 0.0 & -0.65 & 0.65 & 1.0 \end{bmatrix}$
- D. $\boldsymbol{\Sigma} = \begin{bmatrix} 1.0 & -0.65 & -0.65 & 0.0 \\ -0.65 & 1.0 & 0.65 & 0.65 \\ -0.65 & 0.65 & 1.0 & 0.0 \\ 0.0 & 0.65 & 0.0 & 1.0 \end{bmatrix}$
- E. Don't know.

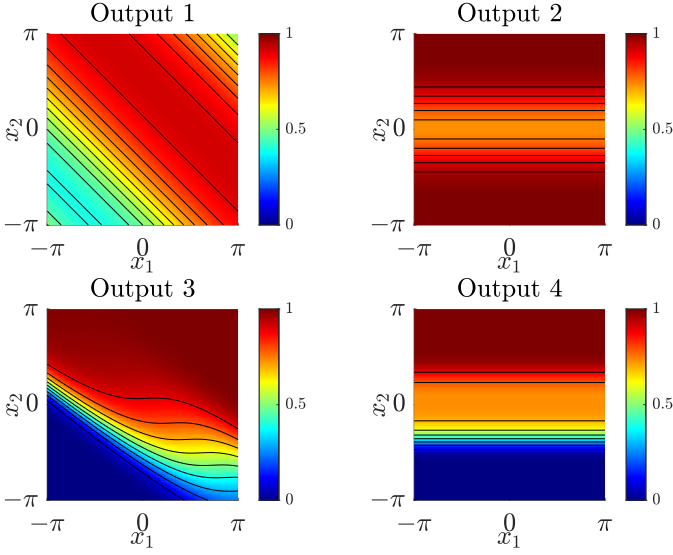


Figure 18: The output of each logistic regression model $p(y = 2 | \mathbf{x})$ is shown as a heatmap with overlaid contour lines.

Question 26. Figure 18 shows the output of four different regularized logistic regression models. We use the binary cross entropy loss to measure the discrepancy between predictions and target values. All the weights, including the bias/intercept term, are equal to 1. The models predict the output based on nonlinear transformations of x_1 and x_2 , while we apply the standard prediction rule $p(y = 2 | \mathbf{x}) = \sigma(\mathbf{w}^\top \hat{\mathbf{x}})$ where $\sigma(z) = \frac{1}{1+\exp(-z)}$ is the logistic sigmoid function. The models are given by:

$$\begin{aligned} M_A: \tilde{\mathbf{x}} &= [x_2^2, 1]^\top \\ M_B: \tilde{\mathbf{x}} &= [\cos(x_1 + x_2), \sin(x_1 + x_2), 1]^\top \\ M_C: \tilde{\mathbf{x}} &= [x_1 + x_2^3, \cos(x_1 + x_2), x_2, 1]^\top \\ M_D: \tilde{\mathbf{x}} &= [x_2^3, 1]^\top \end{aligned}$$

Determine which of the following options is *correct*.

- A. M_A corresponds to Output 3
 M_B corresponds to Output 2
 M_C corresponds to Output 4
 M_D corresponds to Output 1
- B. M_A corresponds to Output 3
 M_B corresponds to Output 2
 M_C corresponds to Output 1
 M_D corresponds to Output 4
- C. M_A corresponds to Output 2
 M_B corresponds to Output 1
 M_C corresponds to Output 3
 M_D corresponds to Output 4
- D. M_A corresponds to Output 4
 M_B corresponds to Output 3
 M_C corresponds to Output 1
 M_D corresponds to Output 2
- E. Don't know.

	$x_3 \in I_1$	$x_3 \in I_2$	$x_3 \in I_3$
Small fish ($y = 1$)	38	30	49
Big fish ($y = 2$)	2	9	31

Table 8: Number of Small fish ($y = 1$) and Big fish ($y = 2$) according to which interval I_1 , I_2 , I_3 , the attribute x_3 belongs to.

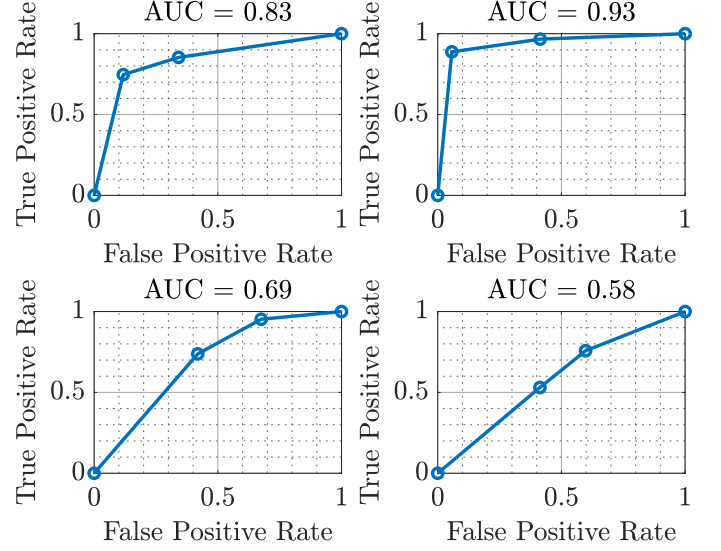


Figure 19: Four different receiver operator characteristic (ROC) curves with area under curve (AUC) values.

Question 27. We consider the attribute x_3 (corresponds to “Height”) of the Fish market dataset. We want to evaluate how well this attribute predicts the class Small fish vs. Big fish based on the data given in Table 8, where the attribute x_3 is partitioned into 3 intervals. For this purpose, we will evaluate the area under the curve (AUC) of the receiver operator characteristic (ROC) created using the attribute x_3 as follows. Given a threshold $\hat{\theta}$, if the $x_3 > \hat{\theta}$ then we classify the point to the *positive class* Big fish ($y = 2$), otherwise the point is classified to the *negative class* Small fish ($y = 1$).

Which one of the ROC curves in Figure 19 corresponds to using x_3 to discriminate between Small fish ($y = 1$) and Big fish ($y = 2$)?

Hint: An exact computation of the fractions is unnecessary; only their relative positions in the plot are required.

- A. The ROC curve having an AUC of 0.83.
- B. The ROC curve having an AUC of 0.69.
- C. The ROC curve having an AUC of 0.58.
- D. The ROC curve having an AUC of 0.93.
- E. Don't know.